Ujaan Das

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Education

University of Warwick — Coventry, United Kingdom
Master of Science in Computer Science — *Sep. 2025 – Sep. 2026*

Hong Kong University of Science and Technology — Hong Kong SAR
Bachelor of Engineering in Computer Engineering — *Sep. 2021 – May 2025*

Northwestern University — Illinois, USA
Exchange Semester — *Jan. 2024 – Apr. 2024*

West Island School — Hong Kong SAR
International Baccalaureate (41/45; HL: Computer Science, Physics, Psychology) — *Sep. 2019 – May 2021*
iGCSEs (English (A), Mathematics (8), Sciences (9); Art (7), Business Studies (B), Computer Science (A), Spanish (A), Global Perspectives (A))* — *Sep. 2017 – May 2019*

Experience

Software Engineering Intern (Part Time) — Jan. 2025 – Aug. 2025
Stellerus Technology — *Hong Kong SAR*

- Re-designed a chatbot execution pipeline using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session
- Standardized message contracts and schema enforcement across the pipeline, eliminating 90% of parsing-related regressions and reducing runtime errors by 30%
- Co-developed the SpaceGPT VLM platform awarded a Gold Medal at the 50th Geneva International Invention Exhibition; engineered the technical demo and automated deployment via Docker and Github Actions

Research Assistant — May 2024 – Sep. 2024
CASTLE Lab, HKUST — *Hong Kong SAR*

- Engineered high-throughput semantic-analysis pipelines in Python and Java to extract semantic signals from 30k+ scraped GitHub repositories to inform research in LLM-assisted testcase generation
- Integrated Tree-sitter to parse multi-million-line codebases into ASTs and CFGs, converting complex source code into lightweight, queryable intermediate representations
- Optimized this parsing logic to cut per-project analysis time by 45%, ultimately increasing usable data yield by 3.2x to support large-scale experiments, slated for publication in ACM ICSE'27

Software Engineering Intern — May 2023 – Aug. 2023
Stellerus Technology — *Hong Kong SAR*

- Developed RESTful APIs using FastAPI and MongoDB to orchestrate multi-user task queues for live satellite data, designed to sustain throughput of 8,000+ requests/sec with <150ms latency and minimal queue times
- Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34%
- Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%

Research Assistant — Jun. 2022 – Aug. 2022
HCI Initiative, HKUST — *Hong Kong SAR*

- Engineered a real-time telemetry and metrics pipeline processing 50K+ events/session, enabling analysis across 4K+ user interactions in large-scale experiments
- Collaborated with post-docs to build a modular React + TypeScript frontend using custom hooks and Tailwind CSS, improving component reusability and streamlining development across multiple researchers
- Contributed to FARPLS, a feature-augmented preference learning system that improved label quality and user engagement without increasing cognitive load (published at ACM/IUI'24)

Projects

MSc Dissertation - Invariants | *C++, Gtest, CMake, Python, Pybind11, Nix* — Github

- Engineering a specification DSL using semantic invariants and refinement types to guide LLM decoding via verifiable runtime constraints
- Architecting a C++ compilation runtime with Python bindings for seamless integration into inference pipelines, enforcing schema compliance with targeted <30ms overhead

Snip | *C++, Termios, Gtest, CMake, Nix* — Github

- Architected an async C++ TUI framework and Vim-style modal editor with a thread-safe, non-blocking event pipeline and background worker pools for deferred execution
- Integrated native LSP tooling for real-time diagnostics and built hermetic, reproducible environments via Nix for cross-platform consistency

yumcha | *Go, C, CGo, Cocoa, Nix* Github

- Engineered a macOS tiling window manager in Go and C using Cocoa/CoreGraphics, implementing an i3-style tree layout engine for dynamic, scriptable window management without disabling SIP

Undergraduate FYP - Follow-Me Robot | *C++, Python, ROS2, Raspberry Pi, Docker* Github

- Developed a real-time UWB tracking system via custom ROS2 Python/C++ nodes, applying multi-rate Kalman and EMA filtering for stable sub-centimeter localization under noisy conditions

oojsite | *Go, HTML, TailwindCSS, Nix* Github

- Built a minimalist static site generator in Go (text/template) that compiles Markdown into composable HTML with reusable components, enabling fast, dependency-free blog generation

TerraViz | *Next.js, React, TypeScript, CesiumJS, TailwindCSS* DevPost

- Built an interactive geospatial platform in Next.js/React for real-time exploration of climate and land-use data
- Developed a 3D CesiumJS visualization layer to render multi-source geospatial data into actionable maps

Skills

Languages: C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS
Frameworks & Libraries: GTest, CMake, FastAPI, Next.js, React.js, Node.js, LangGraph, Pydantic
Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis
Tools & Cloud: Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git